

SymPy Bug Card

Bug 40: Even Powers of Complex Sign Collapse to One

Task. Evaluate an even power of the complex sign at a non-real argument:

$$\text{sign}(1 + i)^2.$$

SymPy's answer.

$$\text{sign}(1 + i)^2 = 1.$$

Correct answer.

$$\text{sign}(1 + i)^2 = \frac{(1 + i)^2}{|1 + i|^2} = i.$$

Explanation. For nonzero complex z , complex sign is the unit direction $z/|z|$. Substituting $z = 1 + i$ gives $(1 + i)^2/2 = i$, so the returned real value 1 is not mathematically equivalent.

Likely root cause. The diagnosis points to `sign._eval_power` in `sympy/functions/elementary/complexes.py`, where an even-integer power of any nonzero `sign` expression is simplified to 1 without requiring a real argument. This is a new instance of a known failure mode.

Artifact (GitHub).

[bug-report/bug-040-even-powers-of-complex-sign-collapse-to-one/](#)